Project Title

Sprint Number
Date

Name	**Email Address**
Name1	WKU email address1
Name2	WKU email address2
Name3	WKU email address3

CS 360
Fall 2024
Project Technical Documentation

Contents

1 Introduction **1**
- 1.1 Project Overview . . . 1
- 1.2 Project Scope . . . 1
- 1.3 Technical Requirements . . . 1
  - 1.3.1 Functional Requirements . . . 1
  - 1.3.2 Non-Functional Requirements . . . 1
- 1.4 Target Hardware Details . . . 2
- 1.5 Software Product Development . . . 2

2 Modeling and Design **2**
- 2.1 System Boundaries . . . 2
  - 2.1.1 Physical . . . 2
  - 2.1.2 Logical . . . 2
- 2.2 Wireframes and Storyboard . . . 2
- 2.3 UML . . . 2
  - 2.3.1 Class Diagrams . . . 2
  - 2.3.2 Use Case Diagrams . . . 2
  - 2.3.3 Use Case Scenarios Developed from Use Case Diagrams (Primary, Secondary) . . . 3
  - 2.3.4 Sequence Diagrams . . . 3
  - 2.3.5 State Diagrams . . . 3
  - 2.3.6 Component Diagrams . . . 3
  - 2.3.7 Deployment Diagrams . . . 3
- 2.4 Version Control . . . 3
- 2.5 Requirements Traceability Table . . . 3
- 2.6 Data Dictionary . . . 3
- 2.7 User Experience . . . 3
  - 2.7.1 Gameplay Diagram . . . 3
  - 2.7.2 Gameplay Objectives . . . 3
  - 2.7.3 User Skillset . . . 3
  - 2.7.4 Gameplay mechanics . . . 3
  - 2.7.5 Gameplay Items . . . 3
  - 2.7.6 Gameplay Challenges . . . 3
  - 2.7.7 Gameplay Menu Screens . . . 4
  - 2.7.8 Gameplay Heads-Up Display . . . 4
  - 2.7.9 Gameplay Art Style . . . 4
  - 2.7.10 Gameplay Audio . . . 4

3 Non-Functional Product Details **4**
- 3.1 Product Security . . . 4
  - 3.1.1 Approach to Security in all Process Steps . . . 4
  - 3.1.2 Security Threat Model . . . 4
  - 3.1.3 Security Levels . . . 4
- 3.2 Product Performance . . . 4
  - 3.2.1 Product Performance Requirements . . . 4
  - 3.2.2 Measurable Performance Objectives . . . 4
  - 3.2.3 Application Workload . . . 4
  - 3.2.4 Hardware and Software Bottlenecks . . . 4
  - 3.2.5 Synthetic Performance Benchmarks . . . 4
  - 3.2.6 Performance Tests . . . 4

4 Software Testing **5**
4.1 Software Testing Plan Template . . . 5
4.2 Unit Testing . . . 5
4.2.1 Source Code Coverage Tests . . . 5
4.2.2 Unit Tests and Results . . . 5
4.3 Integration Testing . . . 5
4.3.1 Integration Tests and Results . . . 5
4.4 System Testing . . . 5
4.4.1 System Tests and Results . . . 5
4.5 Acceptance Testing . . . 5
4.5.1 Acceptance Tests and Results . . . 5

5 Conclusion **5**

6 Appendix **6**
6.1 Software Product Build Instructions . . . 6
6.2 Software Product User Guide . . . 6
6.3 Source Code with Comments . . . 6

List of Figures

1 Introduction

1.1 Project Overview

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Aliquet lectus proin nibh nisl condimentum id. Lorem dolor sed viverra ipsum nunc aliquet. Magna fringilla urna porttitor rhoncus dolor. Bibendum at varius vel pharetra vel turpis nunc eget. Fermentum posuere urna nec tincidunt praesent semper.

Nunc congue nisi vitae suscipit tellus mauris a. Tellus at urna condimentum mattis pellentesque id nibh. Massa tincidunt dui ut ornare lectus. Quisque id diam vel quam elementum. Nunc lobortis mattis aliquam faucibus. Tellus elementum sagittis vitae et. Eget felis eget nunc lobortis mattis aliquam faucibus purus. Risus commodo viverra maecenas accumsan lacus vel facilisis. Nullam vehicula ipsum a arcu cursus vitae. Morbi tristique senectus et netus et malesuada.

Volutpat commodo sed egestas egestas fringilla phasellus. Sed felis eget velit aliquet sagittis id consectetur purus ut. Erat imperdiet sed euismod nisi porta lorem. Ut consequat semper viverra nam. Pharetra diam sit amet nisl suscipit adipiscing bibendum est. Elit ullamcorper dignissim cras tincidunt.

1.2 Project Scope

Text goes here.

1.3 Technical Requirements

1.3.1 Functional Requirements

Mandatory Functional Requirements
Req 1
Req 2
Req 3
Extended Functional Requirements
Ext. Req 1
Ext. Req 2
Ext. Req 3

Text goes here.

1.3.2 Non-Functional Requirements

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Aliquet lectus proin nibh nisl condimentum id. Lorem dolor sed viverra ipsum nunc aliquet. Magna fringilla urna porttitor rhoncus dolor. Bibendum at varius vel pharetra vel turpis nunc eget. Fermentum posuere urna nec tincidunt praesent semper.

Nunc congue nisi vitae suscipit tellus mauris a. Tellus at urna condimentum mattis pellentesque id nibh. Massa tincidunt dui ut ornare lectus. Quisque id diam vel quam elementum. Nunc lobortis mattis aliquam faucibus. Tellus elementum sagittis vitae et. Eget felis eget nunc lobortis mattis aliquam faucibus purus. Risus commodo viverra maecenas accumsan lacus vel facilisis. Nullam vehicula ipsum a arcu cursus vitae. Morbi tristique senectus et netus et malesuada.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Aliquet lectus proin nibh nisl condimentum id. Lorem dolor sed viverra ipsum nunc aliquet. Magna

Mandatory Non-Functional Requirements
Req 1
Req 2
Req 3
Extended Non-Functional Requirements
Ext. Req 1
Ext. Req 2
Ext. Req 3

fringilla urna porttitor rhoncus dolor. Bibendum at varius vel pharetra vel turpis nunc eget. Fermentum posuere urna nec tincidunt praesent semper.

Nunc congue nisi vitae suscipit tellus mauris a. Tellus at urna condimentum mattis pellentesque id nibh. Massa tincidunt dui ut ornare lectus. Quisque id diam vel quam elementum. Nunc lobortis mattis aliquam faucibus. Tellus elementum sagittis vitae et. Eget felis eget nunc lobortis mattis aliquam faucibus purus. Risus commodo viverra maecenas accumsan lacus vel facilisis. Nullam vehicula ipsum a arcu cursus vitae. Morbi tristique senectus et netus et malesuada.

1.4 Target Hardware Details

Text goes here.

1.5 Software Product Development

Text goes here.

2 Modeling and Design

2.1 System Boundaries

2.1.1 Physical

Text goes here.

2.1.2 Logical

Text goes here.

2.2 Wireframes and Storyboard

Text goes here.

2.3 UML

2.3.1 Class Diagrams

Text goes here.

2.3.2 Use Case Diagrams

Text goes here.

2.3.3 Use Case Scenarios Developed from Use Case Diagrams (Primary, Secondary)

Text goes here.

2.3.4 Sequence Diagrams

Text goes here.

2.3.5 State Diagrams

Text goes here.

2.3.6 Component Diagrams

Text goes here.

2.3.7 Deployment Diagrams

Text goes here.

2.4 Version Control

Text goes here.

2.5 Requirements Traceability Table

Text and table goes here.

2.6 Data Dictionary

Text and table goes here.

2.7 User Experience

2.7.1 Gameplay Diagram

Text goes here.

2.7.2 Gameplay Objectives

Text goes here.

2.7.3 User Skillset

Text goes here.

2.7.4 Gameplay mechanics

Text goes here.

2.7.5 Gameplay Items

Text goes here.

2.7.6 Gameplay Challenges

Text goes here.

2.7.7 Gameplay Menu Screens

Text goes here.

2.7.8 Gameplay Heads-Up Display

Text goes here.

2.7.9 Gameplay Art Style

Text goes here.

2.7.10 Gameplay Audio

Text goes here.

3 Non-Functional Product Details

3.1 Product Security

3.1.1 Approach to Security in all Process Steps

Text goes here.

3.1.2 Security Threat Model

Text goes here.

3.1.3 Security Levels

Text goes here

3.2 Product Performance

3.2.1 Product Performance Requirements

Text goes here.

3.2.2 Measurable Performance Objectives

Text goes here.

3.2.3 Application Workload

Text goes here.

3.2.4 Hardware and Software Bottlenecks

Text goes here.

3.2.5 Synthetic Performance Benchmarks

Text goes here.

3.2.6 Performance Tests

Text goes here

4 Software Testing

4.1 Software Testing Plan Template

Test Plan Identifier:
Introduction:
Test item:
Features to test/not to test:
Approach:
Test deliverables:
Item pass/fail criteria:
Environmental needs:
Responsibilities:
Staffing and training needs:
Schedule:
Risks and Mitigation:
Approvals:

4.2 Unit Testing

Text goes here.

4.2.1 Source Code Coverage Tests

Text goes here.

4.2.2 Unit Tests and Results

Text goes here.

4.3 Integration Testing

Text goes here.

4.3.1 Integration Tests and Results

Text goes here.

4.4 System Testing

Text goes here.

4.4.1 System Tests and Results

Text goes here.

4.5 Acceptance Testing

Text goes here.

4.5.1 Acceptance Tests and Results

Text goes here.

5 Conclusion

Text goes here.

6 Appendix

6.1 Software Product Build Instructions

Text goes here.

6.2 Software Product User Guide

Text goes here.

6.3 Source Code with Comments

Text goes here.